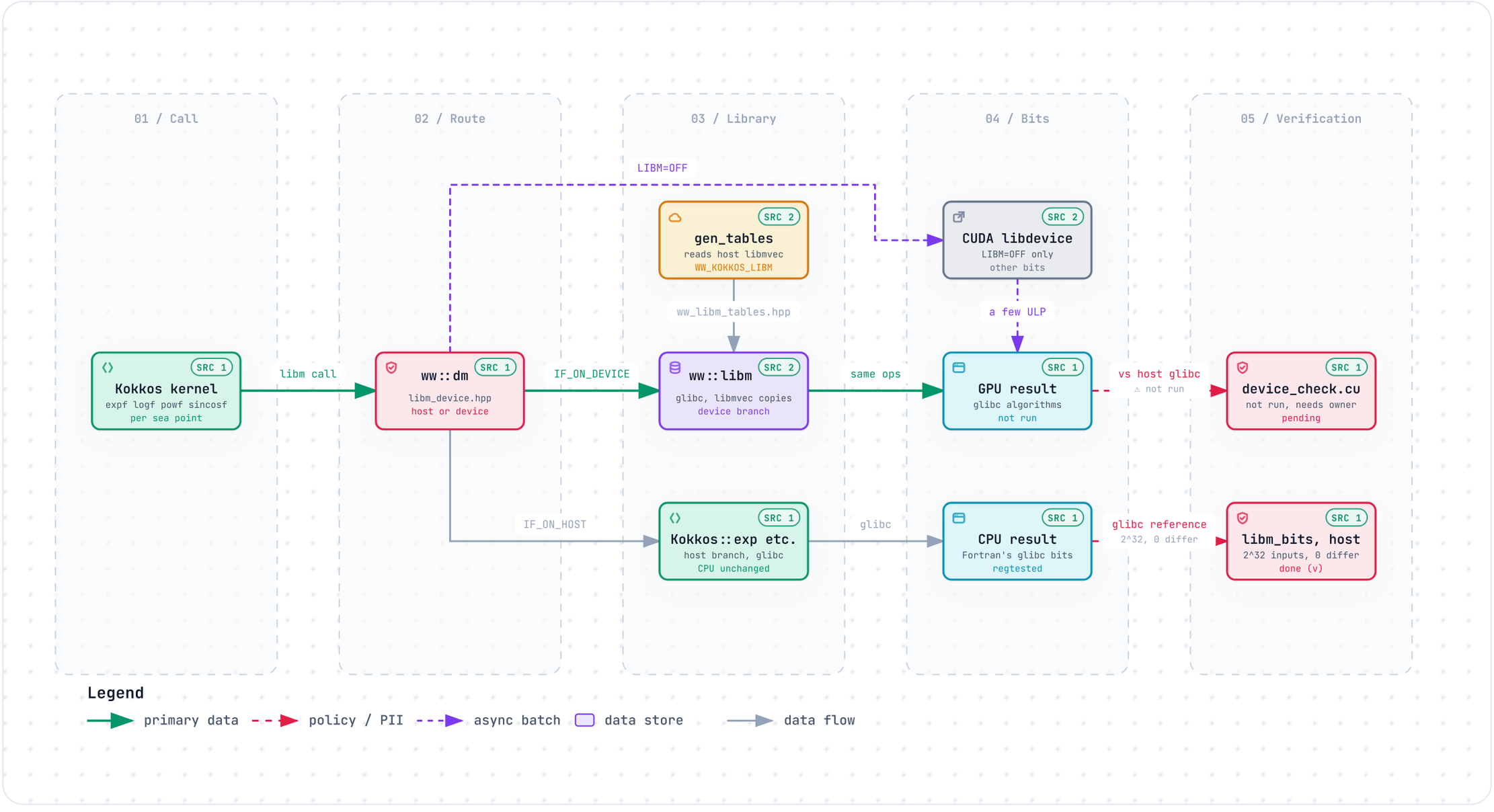
ww3-gpu Device Math Data Flow



Host path

- Each ww::dm wrapper's host branch is the Kokkos call the kernel made before
- CPU objects call glibc as before, so the CPU regtest record stands
- -ffp-contract=off on the C++ (CMakeLists.txt:297-298)

Device path

- CUDA build with WW_KOKKOS_LIBM (default ON) calls ww::libm
- Scalar algorithms: ARM optimized-routines as glibc 2.44 imports them; CORE-MATH for tanhf, log10f, atan2f, tanh
- gen_tables writes libmvec's tables, this host's rcpps and glibc's fma or sse2 choice
- --fmad=false on the device half

Verification

- Host: every float for unary functions, 0 differing (GPU_LIBM_BITS_202610.md)
- SASS: same fused and unfused operation counts as glibc's x86 builds
- Device: scripts/libm_bits.sh --device, under the owner, not run ⚠